Advanced Reinforcement Learning for Bipedal Walking: TQC with D2RL

Anonymous author

Abstract

This paper explores advanced reinforcement learning techniques, specifically Soft Actor-Critic (SAC) with Truncated Quantile Critics (TQC), Deep Dense Reinforcement Learning (D2RL), and an Experience Replay Enhancement (ERE) buffer, applied to OpenAI Gym's Bipedal Walker environments. Our model is able to converge on the normal environment within 140 episodes. While the model shows significant improvements over basic TD3 and TQC models, especially in the hardcore variant, challenges remain in computational demand, parameter tuning, and adapting to dynamic environments.

1 Methodology

This paper details the development of a sophisticated model designed to train a bipedal robot to walk in complex scenarios using advanced reinforcement learning techniques. It specifically investigates the efficacy of combining multiple advanced reinforcement learning techniques—specifically, Soft Actor-Critic (SAC) with Truncated Quantile Critics (TQC), supplemented by Deep Dense Reinforcement Learning (D2RL) and an Enhanced Replay Experience (ERE) buffer—within the challenging environments provided by OpenAI's Gym Bipedal Walker, both normal and hardcore variants. This section details the sequential integration and testing of these methodologies, culminating in a sophisticated model designed to optimize policy learning in complex scenarios.

1.1 Twin Delayed DDPG (TD3)

We initiated our exploration with TD3, aiming to leverage its deterministic policy approach. TD3 introduces two key modifications: it uses a pair of Q-functions to minimize overestimation by taking the minimum of the two estimators, and it adds noise to the target policy to smooth out Q-values and reduce variance [3]. The target value y for the critic update is calculated by:

Given: critic networks Q_{θ_1}, Q_{θ_2}; policy network π_ϕ; target networks $Q_{\theta'_1}$, $Q_{\theta'_2}$, $\pi_{\phi'}$; reward r; discount factor γ; next state s'; clipping range c; noise process ϵ.

$$y = r + \gamma \min_{i=1,2} Q_{\theta'_i}(s', \pi_{\phi'}(s') + \text{clip}(\epsilon, -c, c))$$

The policy π_ϕ is updated by policy gradient methods less frequently, typically every d critic updates.

1.2 Soft Actor Critic (SAC)

Due to challenges with exploration and exploitation balance and high reward variance in TD3, we transitioned to SAC. SAC incorporates entropy regularization into the RL objective, promoting exploration by maximizing both the expected return and the entropy of the policy [4]:

$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}_{(s_t,a_t)\sim\rho_\pi} \left[\gamma \left(r(s_t, a_t) + \alpha\mathcal{H}(\pi(\cdot|s_t))\right)\right] \tag{1}$$

where α is the temperature parameter that controls the trade-off between exploration (entropy) and exploitation (reward).

SAC performed better with increased sampling efficiency, however, possibly due to an overestimation of the Q-function, there was still a high variance in the rewards [3].

Various hyperparameters were experimented with including the Actor and Critics learning rate, batch size,

1.3 Truncated Quantile Critics (TQC)

Persistent high variance and the suspected overestimation of the Q-function led to the adoption of TQC. TQC modifies the critic architecture to estimate a distribution of possible outcomes, which helps in mitigating the overestimation bias typically observed in Q-learning methods [6]:

$$\text{Quantile Loss: } \rho_\tau^\kappa(\delta_{t+1}) = |\tau - \mathbf{1}(\delta_{t+1} < 0)| \frac{\mathcal{H}_\kappa(\delta_{t+1})}{\kappa}, \tag{2}$$

where the Huber loss $\mathcal{H}_\kappa$ is defined as:

$$\mathcal{H}_\kappa(\delta) = \begin{cases} \frac{1}{2}\delta^2 & \text{if } |\delta| \leq \kappa, \\ \kappa(|\delta| - \frac{1}{2}\kappa) & \text{otherwise.} \end{cases} \tag{3}$$

The target update for the quantile regression in TQC is then given by:

$$\delta_{t+1} = r_{t+1} + \gamma \min_{\tau' \in \mathcal{T}} Q_{\text{target}}^{\tau'}(s_{t+1}, a_{t+1}) - Q^\tau(s_t, a_t). \tag{4}$$

The algorithm and parameters were adapted from the Samsung Labs TQC implementation [8]. We further optimized TQC by adjusting the number of quantile approximations, the number of atoms dropped per approximation, and the number of critics. After extensive experimentation, the optimal parameters were determined to be 25 quantile approximations, dropping 2 atoms each, with 5 critics.

1.4 Deep Dense Reinforcement Learning (D2RL)

Building on the enhanced stability from TQC, we aimed to further improve the performance of our model by incorporating D2RL. D2RL enhances the representational capacity of neural networks in RL applications by concatenating outputs from hidden layers with subsequent inputs, promoting deeper feature integration [9]:

$$f(s, a) = \text{ReLU}(W_2 \cdot \text{ReLU}(W_1 \cdot [s; a; h(s, a)] + b_1) + b_2), \tag{5}$$

with $h(s, a)$ representing features from prior layers, allowing for a denser and more expressive model architecture, particularly advantageous in environments with complex input structures.

The code for D2RL was adapted from an existing implementation [7].

1.5 Experience Replay Enhancement (ERE)

Finally, we implemented an Experience Replay Enhancement (ERE) buffer to optimize the use of historical data. By adjusting the sampling probability based on the age of stored experiences, ERE allows the model to prioritize learning from newer, more relevant transitions while still benefiting from older experiences [10]:

$$P(t) = \eta^{\frac{k}{N}}, \tag{6}$$

where η is a decay coefficient, k is the index of sampled transitions, and N is the batch size, allowing for an annealed focus on more recent experiences as learning progresses.

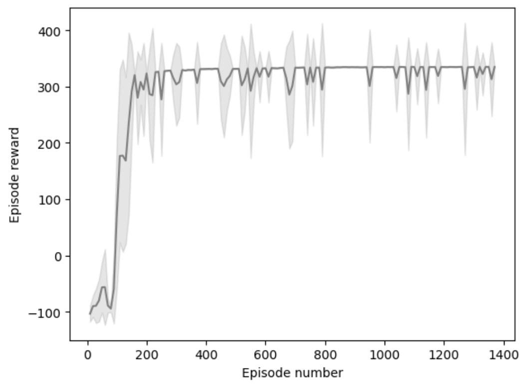

Figure 1: Normal Environment

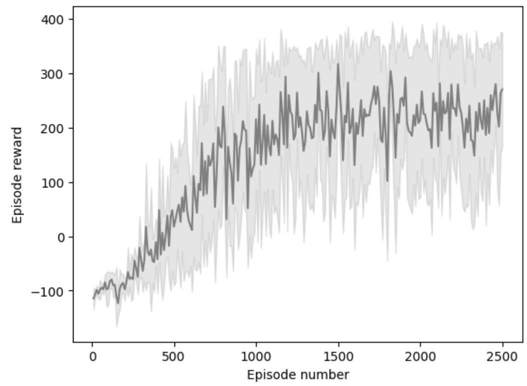

Figure 2: Hardcore environment

2 Convergence Results

2.1 Normal Environment

Our implementation of a SAC model with TQC and D2RL along with an ERE Buffer first solves the normal environment (BipedalWalker-v3) on episode 111 reaching a score of 303.2, and eventually converges from episode 140, as seen in Figure 1. The average reward score over the next 100 episodes, starting from episode 140, is also above 300. On episode 1346, the model reaches the highest reward score of 336.246.

However, the model is not completely stable as there are still occasional dips of rewards falling below 300, due to the model's tendency to keep exploring. Increasing exploitation and reducing exploration can keep the model more stable but it takes longer for the model to find the optimal policy.

2.2 Hardcore Environment

In *BipedalWalkerHardcore-v3*, our model first reaches a score of over 300 at episode 405. As seen in Figure 2, the model never fully converges but does start scoring over 300 quite often after about a 1000 episodes. The model does not do well enough to maintain an average score of 300 over 100 episodes, but can maintain an average score of at least 200 over a span of 100 episodes, with the first occurrence at episode 1082.

The TQC + D2RL + ERE-Buffer model outperformed the basic TQC and TD3 implementation in both, the normal and hardcore environments. The advanced model showed better sample efficiency and policy performance. The initial TD3 model only converged at about 500 episodes while the basic TQC model converged at about 200 episodes.

3 Limitations

Despite the enhancements intended to stabilize learning, models that integrate multiple complex techniques like SAC, TQC, and D2RL can still exhibit high variance in performance across different training runs. This inconsistency can be challenging to diagnose and rectify, particularly when integrating multiple advanced features [5].

Another significant limitation encountered in the application of our model to the Bipedal Walker Hardcore environment is the extensive requirement for hyperparameter tuning. The need for finely tuned parameters is exacerbated by the complexity and unpredictability of the Hardcore environment, which demands a precise balance of model parameters for optimal performance. However, each training run in such a complex environment requires a substantial number of episodes to verify performance improvements, making exhaustive parameter optimization both computationally expensive and time-consuming. This limitation severely restricts the number of experiments that can be realistically conducted within a feasible timeframe, hindering the iterative optimization process that is crucial for refining machine learning models. As a result, the potential for suboptimal performance increases, with the model often failing to converge to the best possible policy due to inadequately explored parameter spaces.

The Bipedal Walker Hardcore environment is characterized by its dynamically changing terrain where the order and nature of obstacles are randomized in each episode. This variability makes each episode effectively a distinct environment, presenting unique challenges that require the model to continuously adapt its strategy. However, our current model architecture, primarily designed for environments with static or predictable dynamics, struggles with such rapid and significant changes. The reliance on historical data from replay buffers, while generally beneficial, may lead the model to make decisions based on previously encountered scenarios that are no longer relevant. This approach can significantly hinder performance as the model fails to update its policy in response to new obstacles and terrain configurations quickly. Moreover, the deep and dense network architectures, such as those introduced in D2RL, while adept at extracting complex patterns from stable environments, tend to overfit to particular types of challenges seen during training. This overfitting diminishes the model's ability to generalize across the varied and unpredictable scenarios presented in subsequent episodes.

Future Work

To address the limitations identified in this study, future research efforts should focus on developing more adaptable learning frameworks that can dynamically respond to rapid changes in complex environments [2]. Implementing meta-learning techniques or modular neural network architectures could potentially enhance the model's adaptability, enabling quicker recalibrations in response to environmental variations. Moreover, improving the management of outdated information in replay buffers—by dynamically adjusting the weighting of older versus newer experiences—could help maintain the learning process's relevance and effectiveness.

Additionally, further exploration into advanced parameter optimization methods, such as using Optuna, could provide a more systematic and efficient approach to fine-tuning the numerous hyperparameters of our model [1]. During this study, we experimented with reward scaling as a technique to potentially enhance learning outcomes. Although initial results did not show significant improvement, there is a strong belief that with sufficient time for more extensive experimentation, this approach could yield better results. Optimizing such parameters through a more thorough and automated search, potentially with Optuna, could unlock higher performance and more robust policies, especially in environments as variable and challenging as the Bipedal Walker Hardcore.

References

[1] Takuya Akiba et al. "Optuna: A Next-generation Hyperparameter Optimization Framework". In: *CoRR* abs/1907.10902 (2019). arXiv: `1907.10902`. URL: `http://arxiv.org/abs/1907.10902`.

[2] Gabriel Dulac-Arnold, Daniel J. Mankowitz, and Todd Hester. "Challenges of Real-World Reinforcement Learning". In: *CoRR* abs/1904.12901 (2019). arXiv: `1904.12901`. URL: `http://arxiv.org/abs/1904.12901`.

[3] Scott Fujimoto, Herke van Hoof, and David Meger. "Addressing Function Approximation Error in Actor-Critic Methods". In: *Proceedings of the 35th International Conference on Machine Learning*. Ed. by Jennifer Dy and Andreas Krause. Vol. 80. Proceedings of Machine Learning Research. PMLR, Oct. 2018, pp. 1587–1596. URL: `https://proceedings.mlr.press/v80/fujimoto18a.html`.

[4] Tuomas Haarnoja et al. "Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor". In: *CoRR* abs/1801.01290 (2018). arXiv: `1801.01290`. URL: `http://arxiv.org/abs/1801.01290`.

[5] Peter Henderson et al. "Deep Reinforcement Learning that Matters". In: *CoRR* abs/1709.06560 (2017). arXiv: `1709.06560`. URL: `http://arxiv.org/abs/1709.06560`.

[6] Arsenii Kuznetsov et al. "Controlling Overestimation Bias with Truncated Mixture of Continuous Distributional Quantile Critics". In: *Proceedings of the 37th International Conference on Machine Learning*. Ed. by Hal Daumé III and Aarti Singh. Vol. 119. Proceedings of Machine Learning Research. PMLR, 13–18 Jul 2020, pp. 5556–5566. URL: `https://proceedings.mlr.press/v119/kuznetsov20a.html`.

[7] Pairlab. *Code for the paper "D2rl: Deep dense architectures for reinforcement learning"*. URL: `https://github.com/pairlab/d2rl`.

[8] SamsungLabs. *Samsung Labs TQC Implementation*. URL: `https://github.com/SamsungLabs/tqc_pytorch/tree/master/tqc`.

[9] Samarth Sinha et al. "D2RL: Deep Dense Architectures in Reinforcement Learning". In: *CoRR* abs/2010.09163 (2020). arXiv: `2010.09163`. URL: `https://arxiv.org/abs/2010.09163`.

[10] Che Wang and Keith W. Ross. "Boosting Soft Actor-Critic: Emphasizing Recent Experience without Forgetting the Past". In: *CoRR* abs/1906.04009 (2019). arXiv: `1906.04009`. URL: `http://arxiv.org/abs/1906.04009`.